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Communication-efficient and privacy-preserving federated learning for medical image classification in multi-institutional edge computing

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Abstract

To effectively treat patients, health care providers must be able to detect diseases early and diagnose them accurately. Deep learning and computer vision have recently enhanced the diagnostic accuracy of skin cancer through image-classification models. However, the centralized learning (CL) method is problematic in terms of data privacy because of the constraint of transferring substantial volumes of data to a central server. In this study, we investigate edge intelligence training in a multi-institutional edge environment to address data privacy, machine learning training delay, and training data limitations at each healthcare center. We considered skin cancer image classification as our use case in light of skin cancer diagnosis and assisting health care experts. Initially, we focused on the training delays induced by communication and computational latency. The edge-average federated learning (Edge-Avg) method improved the training latency by 24% and 47% compared with the federated learning (Fed-Avg) and CL approaches, respectively. This indicates reduced processing time, which is particularly crucial in the field of medical diagnostics. In terms of accuracy, our Edge-Avg approach surpassed the Fed-Avg method by achieving a better accuracy of 93.94% compared to the traditional method in the classification of skin cancer images into benign or malignant categories. This study provides a low-latency and accurate solution to assist medical professionals in assessing the status of suspected skin cancer cases.

Keywords Edge intelligence, Federated learning, Medical image classification, Edge computing, Deep learning

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Introduction

Medical image classification is a critical application in healthcare, as it assists healthcare professionals to diagnose medical diseases accurately and efficiently. However, analysing medical images can be a lengthy process and demands significant computational resources. Artificial intelligence (AI) specific algorithms can automate this process and help healthcare professionals make more accurate and faster diagnoses. Edge intelligence involves the utilization of AI algorithms on edge servers (i.e., in cases of sufficient computation capability). This allows for the analysis of data and decision-making in real time [1]. It enables the computation to be performed locally, reducing the need for data transmission to a cloud computing server, which can be slow and consumes a large amount of network bandwidth, and holds great promise for improving public health. Edge Intelligence can bring this automation to the point of care, providing real-time analysis and diagnosis of medical conditions.

Although edge computing servers have emerged as a powerful platform for processing data and performing AI algorithms at the network edge, they may face certain limitations that prevent them from being effectively processed [2]. One of the primary challenges of edge-computing servers is their limited computational resources. Edge servers usually have lower computational capabilities than cloud servers, so they have a hard time processing large amounts of data or performing complex AI algorithms. This can result from an increase in processing time or the need to upload computations to more powerful servers, which can introduce more latency and reduce the system efficiency. One more obstacle is the restricted access to datasets. AI algorithms, especially in the field of medical image classification, depend on extensive data for efficient training and precise outcomes. This may result in unreliable or inadequate predictions, potentially impacting critical healthcare tasks like skin cancer diagnosis.

Data privacy is a significant concern in several big data-related use cases, where vast amounts of personal and sensitive data are being generated every day [3]. The CL approach to training the model involves the collection of data on a central cloud server, which can pose privacy risks because the data are vulnerable to attacks or breaches. However, medical data are highly sensitive and must be kept confidential to protect patient privacy. Multi-access edge computing can enable the training of AI models on decentralized edge servers while ensuring that the data remain on the edge and are not transferred to a central server or even peer edges. This mitigates the risk of a data breach. In addition, using federated learning approaches in a multi-access edge computing environment can overcome this challenge by enabling the training of AI models on decentralized edge servers while maintaining the data on the device. Multi-access edge computing allows multiple edge servers to collaborate to train the AI models. This collaborative approach can enhance the performance of AI models by integrating the knowledge of multiple edge servers, resulting in a better AI algorithm performance. However, providing edge intelligence for medical image classification in multi-access edge computing poses several challenges. One of the significant challenges is ensuring that the AI models are accurate and reliable. Moreover, federated learning can address the reliability challenges in multi-access edge computing. Network failures and outages, which both originate data at the network edge and process that data locally, may affect AI models by their very nature. This challenge can be mitigated through federated learning, by training models across many edge servers and making them more reliable. On the other hand, federated learning can decrease data homogeneity and balance of geo-distributed edge servers on which models are trained. This can cause model bias, or overfitting, etc.—all of which will reduce the quality of the AI models. In a multi-access edge computing environment, providing edge intelligence capabilities for medical image classification can provide significant benefits in healthcare diagnosis (e.g., skin cancer diagnosis), such as reducing the AI model training time, faster analysis, improved accuracy, and better privacy in patient data.

This research investigates edge intelligence training in multi-access edge environments to address each healthcare centre's data privacy, model training latency, and training data limitations. We consider skin cancer image classification as our use case in light of the skin cancer diagnosis, assisting health care experts.

Main contribution of this study are as follows:

- To reduce overall training time, we proposed a low-latency distributed learning mechanism that optimizes the influencing parameters. To accomplish this, we investigated the trade-off between communication latency and computation latency.
- Edge intelligence facilitates skin cancer classification and detection through an edge optimized federated learning approach.

The subsequent sections of this paper are organized as follows: Background information is presented in Sect. " Background", followed by a thorough review of the literature in Sect. " Literature Review". The system model and problem statements are detailed in Sect. " System Model and Problem Statement". Sect. " Learning algorithm" delves into latency and learning algorithms. A comparative analysis with benchmark models is

conducted in Sect. " Evaluation", with conclusions and discussions included in Sect. " Conclusion".

Background and Literature Review

Background

Edge intelligence

In recent years, there has been a significant growth in the Internet of Things (IoT) and intelligent devices, leading to a proliferation of data at the peripheries of networks. Edge servers, which include cameras, smart appliances, and wearable devices, are near the data source. These data may be insightful and useful, but moving them all to the cloud for processing and analysis is not always feasible. By locally training data at the network edge, edge intelligence minimizes latency, enhances performance, and delivers quicker responses. The applications of edge intelligence are wide-ranging and diverse. In industries, edge intelligence can be used to monitor machines and optimize processes, whereas in healthcare, it may be used to oversee patients and provide immediate analysis to medical professionals [4].

Federated learning

This is an AI technique that enables distributed model training without requiring a central data storage location. This approach is particularly relevant in areas such as healthcare care, where strict privacy regulations and confidential patient data make central data storage impractical [5]. Federated learning allows multiple healthcare institutions to collaborate and share AI models, while maintaining patient privacy. This technique ensures that each institution retains control over its data and only model parameters are exchanged between institutions. This enables larger and more diverse datasets to train the models. A common way to combine model updates from different devices or servers in federated learning is federated averaging [6]. This technique involves computing the average local model weights updated on each edge server during the training phase. This technique is to train a local model in every edge server using its data alone and then transfer these trained models once learning (model size) in the federated information system, FIS [7, 8]. It then calculates the weighted average of the model weights and releases its updated global model to each device or server. This cycle was continued until the model at a global level converges. Federated learning represents an effective method for developing machine learning models utilizing decentralized data sources, as illustrated in Fig. 1.

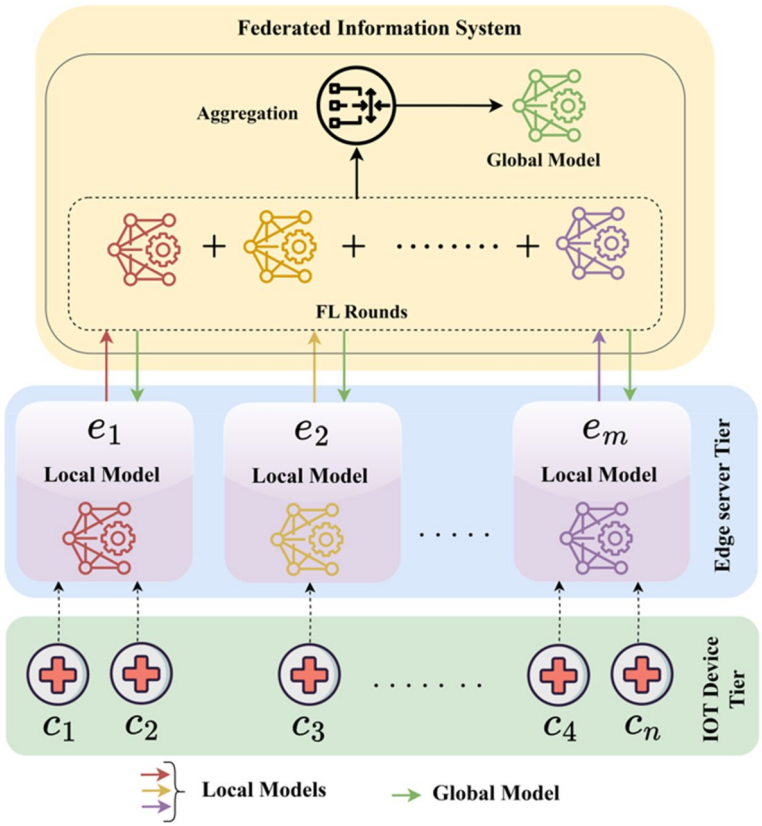


Fig. 1 Federated learning

Literature review

Researchers have been working on several investigations over the past few years to find better and more accurate ways to detect skin cancer (SCD). The authors in [9] established a federated learning system that decentralizes data and safeguards the security and privacy of patient records. They used a dataset with over 10,000 images and attained an initial accuracy rate of 79%. However, by implementing data balancing techniques, the classifier's performance improved significantly, achieving a classification accuracy of 95%. In [10], researchers successfully classified skin lesions from dermoscopy images via a Deep CNN and transfer learning, with an accuracy of 80.3%. In [11], Deep CNNs were applied for skin lesion recognition. Moreover, [12] demonstrated that a CNN model incorporating residual learning and multiscale contextual information integration achieved an 85.5% accuracy in melanoma detection within the ISBI 2016 skin lesion analysis. The dataset was segmented by both the CNN and VGG16 models into non-melanocytic skin diseases and benign or malignant skin diseases categories. Furthermore, the International Skin Imaging Collaboration (ISIC) 2017 challenge dataset was utilized. Our proposed techniques produced promising outcomes, demonstrating an average segmentation value of 0.507. The CNN model obtained AUC values of 0.76, while the VGG16 model achieved 0.86. In [13], the HAM10000 dataset served as a training ground for pretrained neural network architectures to recognize seven types of skin lesions. The resizing and data augmentation applied to the ResNet50 model showed the highest accuracy, which was further boosted to 84% using an ensemble approach combining the ResNet50, VGG16, and DenseNet models. In [14, 15], the researchers developed a skin cancer diagnostic system using transfer learning models such as ResNet152 and InceptionV3. They achieved accuracy rates of 90.4% and 85.8% for diagnosing melanoma, respectively. In [16], the discrete wavelet transformation technique was used to derive attributes from skin cancer images, achieving an accuracy rate of 84% using a support vector machine classification. Another study [17] used ESRGAN techniques to improve the skin lesion image quality and considered multiple transfer learning models, achieving an accuracy rate of 85.7%. In [18, 19] used a sharpening filter and hair removal method to preprocess the images. Deep learning models were used to classify lesion moles, resulting in an accuracy of 86%. In [20], a hybrid deep-learning technique combining convolutional neural networks and a support vector machine classifier achieved a success rate of 90.9%. In [21], a blockchain-federated learning framework was proposed for brain tumor segmentation, focusing on ensuring data privacy, fostering trust and enabling collaboration. The framework addresses challenges, such as integration and regulatory compliance, aiming to improve diagnostics and healthcare outcomes.

To leverage unlabeled data, [22] integrated active learning (AL) into federated learning, proposing an AL-based federated learning framework. Studies of natural disasters and waste classification have shown that AL can be used in both centralized and federated learning, producing results that are similar to those obtained by hand annotating with fewer samples and no human annotators. A different study used the cancer denome atlas public repository dataset to show how the network of healthcare providers works in a distributed system [23]. The authors [24] proposed the idea of a federated robust decoupling (FED-ROD) framework to help find a good balance between a model's general and personalized performance in federated learning. FED-ROD divides the model's tasks into two prediction tasks, considering robust loss functions for general prediction and a lightweight adaptive module for tailored prediction. This strategy allows for the simultaneous achievement of cutting-edge performance in both objectives, successfully bridging the gap between them within the federated learning structure. In another article, the authors [25] proposed a decentralized and collaborative framework that enables clinicians to exploit important private data while maintaining anonymity. Despite the decentralized structure of the data and the obstacles caused by non-IID and unbalanced distributions, the federated learning strategy is robust and provides comparable results to classic centralized learning approaches. Through comprehensive experiments and comparisons, the authors underline the necessity and effectiveness of their proposed technique, particularly in important scenarios, such as the COVID-19 pandemic. Finally, [26] suggested a certain model aggregation method to fix issues with image quality and differences in processing power between vehicle clients when it comes to AI applications for smart connected cars (ICV). The method involves selecting the best"fine" local deep neural network (DNN) models to send to the central server. The distributed framework is based on two-dimensional contract theory to overcome knowledge asymmetry and solve the problem with a greedy algorithm. Tests on the MNIST and Belgium TSC datasets show that this method is more accurate, faster, and more useful at the central server than baseline federated averaging methods.

In summary, taken together these studies shed light on the effectiveness of deep learning Collectively, these studies serve as evidence for the utility of deep learning approaches in skin cancer classification and detection. It has been shown in several studies that by leveraging preprocessing techniques as well as transfer learning and data augmentation we can improve model accuracy. The

Table 1 Comparison of selected related works

Reference	*Fed-Avg*	*Edge-Avg*	*Latency*
[22]	✓	x	x
[23]	✓	x	x
[24]	✓	x	x
[25]	✓	x	x
[26]	✓	x	✓
Our work	x	✓	✓

patient data can be kept private with federated learning in decentralized datasets. In this context, to solve the problems related to training latency and lack of training dataset, we conduct a study about federated learning method across multiple devices edge environments—as well as introducing an Edge-Avg method. Table 1 shows a comparison between different aggregation methods (i.e., Fed-Avg and Edge-Avg) for various related works, including ours, in terms of latency as a performance metric. The results of the study in Sect. " Evaluation" provide evidence for the efficacy of the approach.

System model and problem statement

We envisage a two-tier IoT multi-access edge-computing environment. Let $C = c_1, c_2, ..c_n$ denote the group of clients at the IoT tier, where each client $1 \leq i \leq n$, represents an IoT device. Each client c_i is a source of medical images that are collected by users at related healthcare centers. We consider $E = \{e_1, e_2, ...e_m\}$ to be the set of geo-distributed edge servers in the form of multi-access edge computing. We assume that each edge server e_j, $1 \leq j \leq m$, is situated in a specific medical healthcare system. In this research, we focus on automating SCD within healthcare environments, where each client is responsible for sampling skin images and uploading them to the edge server situated under its service coverage area. In addition, we assume that, owing to computational resource limitations, clients may not be able to fully process medical images. To address this issue, we will upload medical images to related edge servers for processing.

As shown in Fig. 2, each edge server e_j can run AI algorithms associated with medical image processing and uploaded from clients. According to the proposed system model, clients that face computational resource limitations can upload their medical images (i.e., raw data) to the related AI-enabled edge servers for processing. Each edge server was responsible for training the local models using the uploaded data. By using the Fed-Avg approach, the locally trained models are shared among the participating edge servers instead of transferring the raw dataset among the edge servers.

(i.e., to support data privacy). Although the data privacy issue will be solved, we still face computation and communication delays at the edge servers and in between. Basically, factors such as the model's complexity, size of the training dataset, hardware, and optimization strategies can affect the delay of the AI model during its training. Distributed training and hardware accelerators help speed up AI model training by reducing training latency, improving the model's performance (i.e., adapting to changes in the data), and enhancing the model's responsiveness to real-world events.

Let d_{ij} be the size of the raw data, specifically the image size, transmitted from client c_i to the edge server e_j. We define l_j as the size of the global model transferred from the FIS to edge server e_j. However, we consider $l_{j'}$ as the size of the aggregate model at edge server e_j (i.e., the trained and aggregated models at e_j), which is ready to be sent to the FIS. Let b_{ij}^{α} and b_j^{β} be the link capacity (Mbps) from client c_i to edge server e_j and the link capacity between the FIS and edge server e_j, respectively. Without loss of overview, we consider the same link capacity to transfer data from the FIS to the edge server, and vice versa. Therefore, the overall communication latency, t_{γ}^{j} can be calculated using Eq. (1).

$$t_{\gamma}^{j} = \frac{d_{ij}}{b_{ij}^{\alpha}} + \frac{l_j + l_{j'}}{b_j^{\beta}} \tag{1}$$

The efficiency and capability of GPUs in handling large and complex model training tasks makes them a preferred choice for neural network computations over CPUs. The local computation process involves the training time for the data uploaded by clients. We propose modifying the training model and integrating a delay function to capture latency. Let η_j be the number of epochs of the local training system at edge server e_j. In e_j, we considered t_j to be the time per epoch associated with the local training system. With respect to the aggregation of the training model (at each edge server), we calculated the size of the aggregated model l_j on the sum of the trained local models $\sum_{i=1}^{n} l_i$ by the number of clients at the edge server e_j. Therefore, the computation latency t_{ω}^{j} associated with the local training system at the edge server e_j can be calculated using Eq. (2).

$$t_{\omega}^{j} = \frac{\eta_j t_j + \sum_{i=1}^{n} l_i}{n} \tag{2}$$

The effectiveness of the suggested approach is assessed with regard to the comprehensive computational latency (i.e., associated with local training at the edge) and the communication latency associated with updating the locally trained models within one communication round. During training, the updated models are exchanged between the FIS and the edge servers. Therefore, the overall training latency is calculated using Eq. (3).

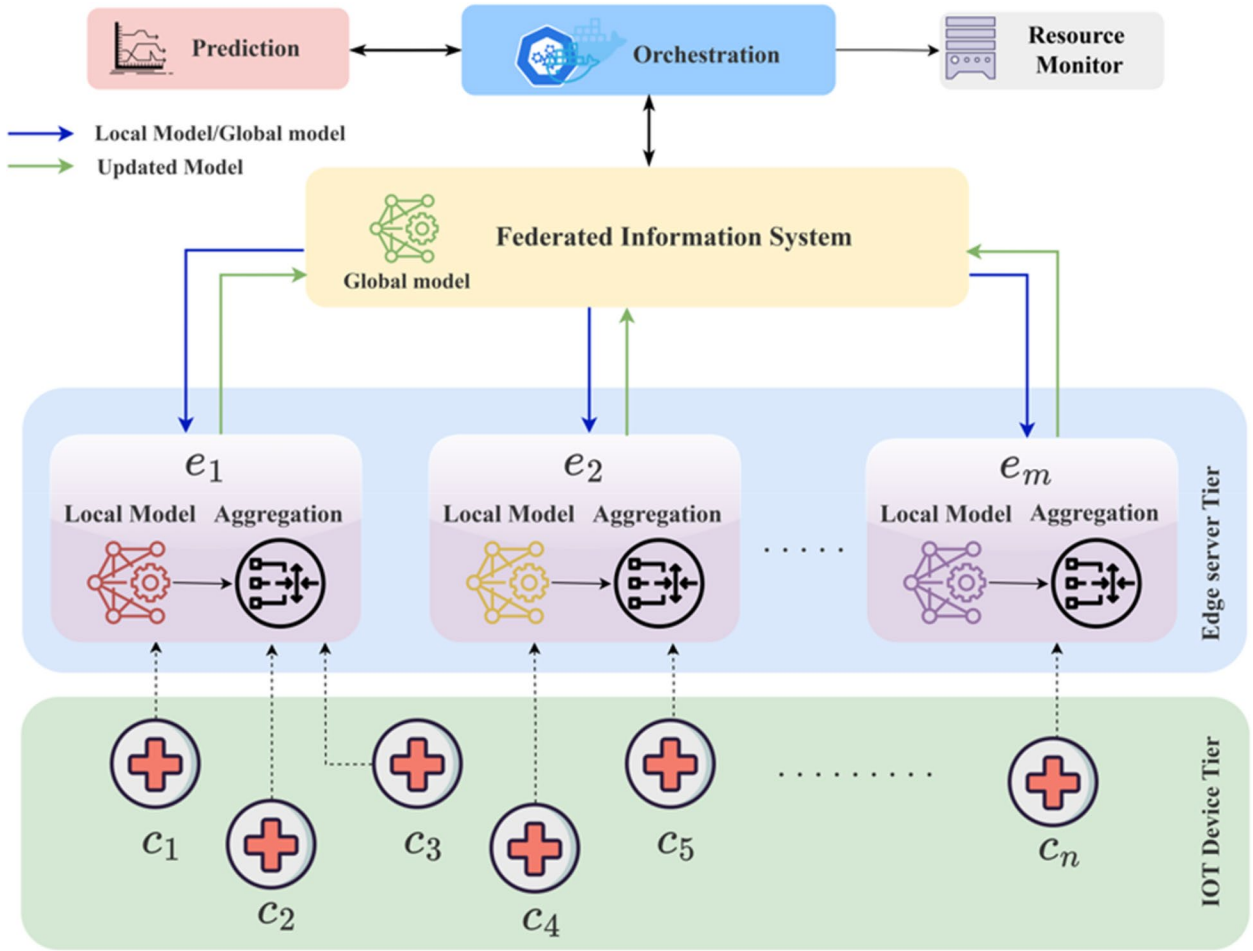


Fig. 2 System model

$$L = t_{\gamma}^{j} + t_{\omega}^{j} \tag{3}$$

Furthermore, we analyzed the performance of proposed approaches using statistical measures [27]. Mean latency ξ provides the average latency in Eq. (4), where L_k is the latency of the kth approach and N is the total number of approaches. The standard deviation Δ captures the variation in latency values, reflecting the consistency of Eq. (5). The coefficient of variation Θ shows the variability as a percentage of the mean for the dataset comparisons in Eq. (6). The improvement percentage Ψ assesses the performance advantage of the proposed approach over others by comparing latencies L_o (FedAvg and CL) and L_a (Edge-Avg) in Eq. (7). Together, these measures provide a clear view of efficiency and reliability.

$$\xi = \frac{\sum_{k=1}^{N} L_k}{N} \tag{4}$$

$$\Delta = \sqrt{\frac{\sum_{k=1}^{N} (L_k - \xi)^2}{N}} \tag{5}$$

$$\Theta = \frac{\Delta}{\xi} \times 100 \tag{6}$$

$$\Psi = \frac{L_o - L_a}{L_o} \times 100 \tag{7}$$

Data-augmentation

In addressing the challenges of data scarcity on edge servers and stringent privacy requirements in healthcare, past discussions, such as [28] and [29], highlight the limitations of available datasets and the importance of data privacy. To address this issue, we used data augmentation to improve the quality of the training datasets. This involves implementing methods such as rescaling, rotation, zooming, width shifting, and horizontal flipping. which is particularly important when there are

limitations in accessing actual patient data. In addition, we incorporated differential privacy and federated learning to comply with privacy regulations, which restrict the sharing of data among peer edge servers and enhance the training data efficiently, as exemplified by the *ImageDataGenerator* function. Implementing these measures is essential for reducing overfitting, which, in turn, enhances the accuracy of predictive models and enables the deployment of healthcare solutions directly on edge servers at the point of care. Additionally, these measures ensure compliance with data privacy regulations. This approach highlights our dedication to promoting the development of healthcare technology by employing responsible data and robust methodological methods.

Transfer learning

This highly effective approach enables the utilization of knowledge from extensive and comprehensive datasets in situations where there are limited data available, such as medical edge servers. This approach improves the precision of diagnosis and predictive analysis, without the need for large amounts of local data. Additionally, by incorporating differential privacy into the transfer-learning framework, we guarantee that our techniques adhere to strict privacy regulations, thereby protecting data confidentiality. The authors of [30] discussed the significance of transfer learning in deep neural networks and provided strategies for mitigating overfitting in situations with limited data. This additional example demonstrates the benefits of transfer learning when a pre-trained model functions as a feature extractor, leveraging ImageNet's extensive generic features to greatly improve its performance. This approach not only improves the quality of diagnostic tools but also enhances the efficient utilization of computational resources and increases the training efficiency. Using *ImageNet* weights in our adaptation provides a better performance than models trained from scratch. This proves that transfer learning is an important way to use advanced healthcare solutions, while still protecting privacy.

DensNet121

Our study focused on investigating the capabilities of DenseNet121, a deep learning model first introduced by [31]. This model presents a unique architecture that is specifically designed for image recognition tasks. DenseNet121 is notable for its densely connected layers that create direct connections between every layer in a feed-forward manner. This architectural design improves the transmission of information and gradients throughout the network, resulting in enhanced feature reuse and a substantial decrease in necessary parameters. This simplifies the model and lowers computational requirements. This procedure not only facilitates the expansion of the network but also improves efficiency by enabling the processing of larger and more complex datasets without requiring additional computational resources. DenseNet121 has been proven to be highly effective for medical imaging. Its strong feature-propagation capabilities effectively address the issue of vanishing gradients, thereby making it a potent tool for disease diagnosis based on medical images. In our research, we performed experiments by decreasing the number of trainable parameters in the model to 1,00,354, a significantly smaller number than the total of 7,137,858. We observed that this reduction in the parameters resulted in improved performance. This finding emphasizes the potential of the model to support the development of new diagnostic expertise that combines fast analysis with a minimal computational burden. This can significantly enhance patient care and improve health outcomes.

Learning algorithm

Fundamentally, CL indicates the process of consolidating all the training data into a single location to build a model, which can be a time-intensive task, particularly for extensive datasets. The Fed-Avg method distributes the training data among multiple servers. The related clients within each server's service coverage area send their datasets for local training. FIS builds a global model using server-trained local models. This approach resulted in reduced training time compared to the CL approach. Conversely, the Edge-Avg method executes the aggregation process at the edge server. This reduces the need for data transmission and further reduces the training delays.

Algorithm 1 shows the latency calculation process for an edge-based distributed learning approach, as described in lines $10-14$. This approach was used to train models in a distributed environment. The algorithm reads input parameters, such as client and global model, initializes all necessary variables, and enters a loop that continues until all clients have completed their training. For each communication round, the algorithm reads the data from the clients and applies the optimizer, batch size, epochs, and learning rate to obtain a locally trained model. The algorithm consolidates the models provided by clients on the edge server and transmits the revised local model size to the FIS. It is essential to emphasize that each edge server executed the medical image processing model. This strategy facilitates effective computations and processing among numerous clients through distributed learning.

Algorithm 1 Calculating the latency of edge-average federated learning

Input:
c_i: clients
l_i: global model
initialize:
l_r= 5e-4 : learning rate
b_s = 15 : batch size
η_j= 20 : epochs
η_c= 20 : communication rounds
while c_n> 0 **do**
  for round in range n_c **do**
    for client in range c_i**do**
      $e_j \leftarrow (c_i, l_i)$
      $l_i \leftarrow$ local_model_train(b_s, η_j, l_r)
      $l_j' = \frac{\sum_{i=1}^{n} l_i}{n}$
      $Update_FIS(l_j')$
    end for
  end for
end while

Algorithm 2 shows the procedure for the edge-based distributed federated learning. This enables efficient processing and training of AI models for clients. The algorithm requires input parameters, such as clients and global model size. It initializes all parameters and enters a loop that iterates until all communication rounds are completed. During each round, the algorithm reads data from clients and augments it using various techniques such as rescaling, rotation, zoom, width-shift, and horizontal flip. An edge server then processes the augmented data using convolution and pooling techniques (i.e., the DensNet121 model), and locally saves the training data (i.e., the local model). We computed the aggregate (i.e., updated model) by adding the local model and all clients, and repeating the procedure for a certain number of communication rounds. Here, *Edge Data* function returns a pre-processing result from clients, *Edge Server* processes the pre-processing, and the *Model Aggregation* function is used to create an updated model based on aggregating local models from all clients. This algorithm enables the training of AI models in a distributed environment by using techniques such as data augmentation, convolution, and pooling. This distributed learning approach has the potential to drive significant advances in AI and IoT technologies, resulting in smarter and more responsive systems that can adapt to real-world environments.

Algorithm 2 Edge-average federated learning

input
c_i: clients
l_j: global model
initialize:
l_r = 5e-4 : learning rate
b_s = 15 : batch size
η_j = 20 : epochs
n_c = 20 : communication round
while n_c > 0 **do**
  for round in range n_c **do**
    $p_i = Edge_Data(c_i, b_s)$
    $l_i = Edge_Server((p_i, l_j, \eta_j, l_r))$
    l_j' = model_aggregation(l_i, c_i)
    $Update_FIS(l_j')$
  end for
end while
procedure edge_data(c_i, b_s):
pre-processing p_i = [ ]
while n_c > 0 **do**
  P^n_{matrix} , F^n_{matrix} = read the dataset from c_i
  $ImageDataGenerator$ = [re-scale, rotation, zoom, width-shift, horizontal flip]
  P^n_{matrix} , P^n_{matrix} = $ImageDataGenerator$ (b_s, class, image size)
end while
return pre-processing
end procedure
procedure (edge_server (c_i, l_j, η_j , l_r)):
local_trained models l_i = []
for clients in range c_i **do**
  for epoch in range η_j **do** // training process (i.e., local model)
    P^n_{matrix} ← convolution_and_pooling (c_i, l_r)
    $l_i \leftarrow P^n_{matrix}$, P^n_{matrix}
  end for
end for
end procedure
procedure model_aggregation (l_i, c_i): // The aggregation is based on summation of the local model and number of clients at the edge server e_j
$l_j' = \frac{\sum_{i=1}^{n} l_i}{n}$
end procedure

Evaluation

This section evaluates the efficacy of our method and compares it with those of other authors. We begin by introducing the dataset and our experimental setup. Next, we analyzed the performance and evaluated different methods.

Dataset

We considered the ISIC-2019 [32], HAM10000 [33] and ISIC-2020 [34] datasets, which are highly valuable for skin cancer classification and model training owing to their extensive and diverse collection of tumor images. Expert-labeled annotations ensure high accuracy and reliability, which are essential for developing effective machine learning models. These datasets include a wide variety of cases, ranging from simple to complex, offering rich features that enhance the models' ability to perform in real-world scenarios. Specifically, the ISIC-2019

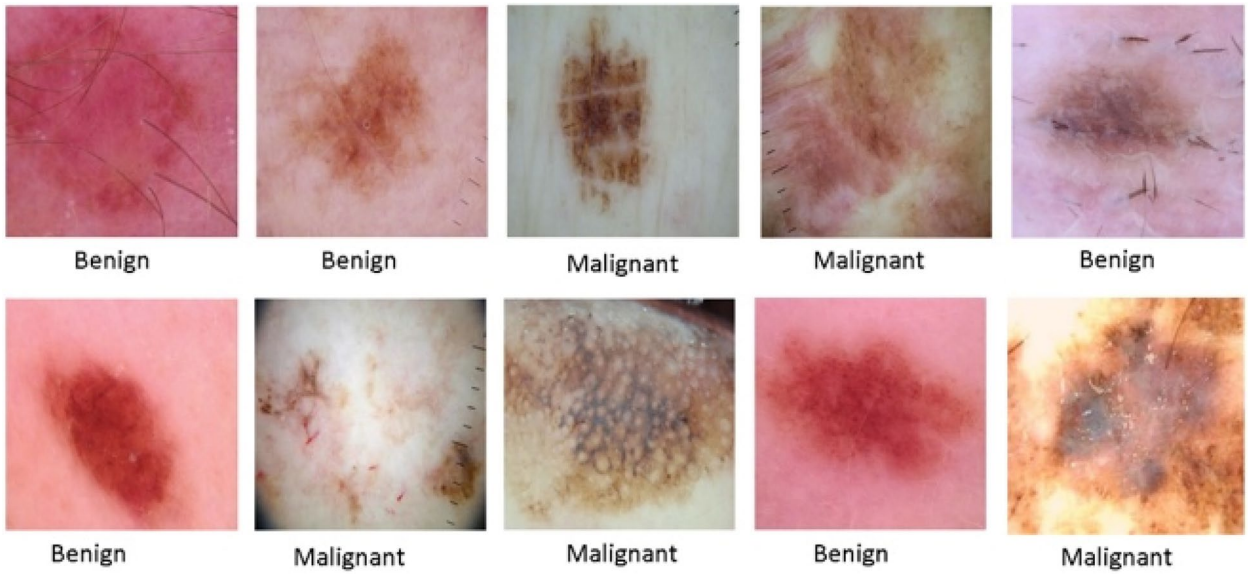


Fig. 3 Benign and malignant cancer dataset

dataset comprises 3,297 dermatological images, with 1,800 representing benign tumors and 1,497 representing malignant tumors, as shown in Fig. 3. Each image was resized to 224×224 pixels to maintain sufficient detail for advanced image processing algorithms. The dataset was partitioned into two subsets, with 80% (approximately 2,638 images) used for training the machine learning models and the remaining 20% (approximately 659 images) designated for validation, enabling us to assess training efficiency and validate model performance effectively.

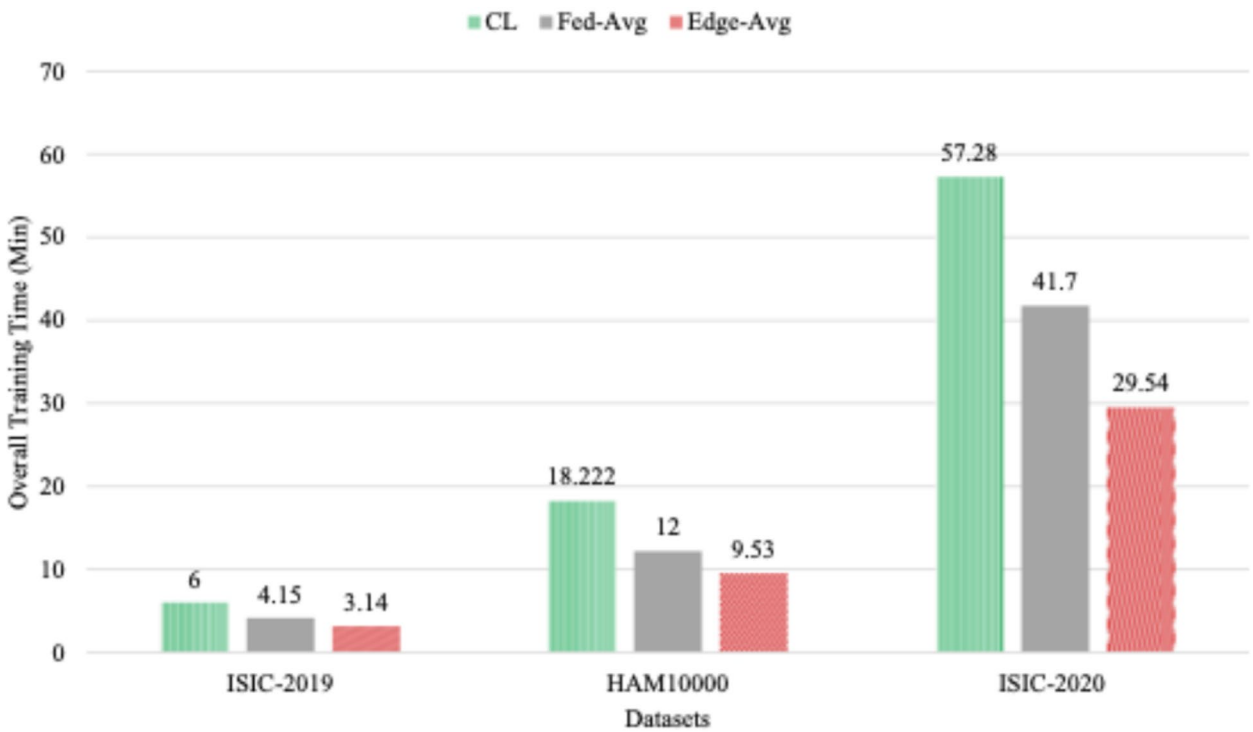


Fig. 4 Overall training time vs different datasets

Experimental setting

An experimental environment was created to assess the effectiveness of the edge averaged federated learning image classification system. We conducted an evaluation using the Google Colab Pro + server with GPU capabilities. The dataset, which consisted of high-resolution skin cancer images from the ISIC-2019, HAM10000 and ISIC-2020 collections, was divided into four separate clients to replicate a decentralized learning situation. We used Python 3 in combination with TensorFlow 2.13 to execute the DensNet121 deep learning models for every client. This solution utilizes Tensor Flow's powerful features for efficient computation and reliable model training across edge servers. To comprehensively assess the efficiency of our Edge-Avg system, we focused on four important metrics: the training latency, accuracy, and loss at various learning rates. By including the DensNet121 model, which is renowned for its extensive layers and intricate structure, we were able to comprehensively evaluate the resilience of our Edge-Avg method in real-world scenarios. The goal was to determine the temporal efficacy of the proposed distributed system by calculating the training delay. Furthermore, examining the accuracy and loss of the model at different learning rates provided valuable information regarding the most effective configurations for achieving excellent classification results. This enabled us to optimize the models for both efficiency and effectiveness in an Edge-Avg environment.

The effect of learning techniques on training delay

We used three main datasets, ISIC-2019, HAM10000 and ISIC-2020, to examine the training delays resulting from three different learning methods: Edge-Avg, Fed-Avg, and CL. As shown in Fig. 4, The Edge-Avg method involves training local models on edge servers and subsequently averaging these models to update the global model on the same servers. This approach reduces latency by efficiently aggregating the local input and updating the model. In contrast, the Fed-Avg method involves training models on multiple edge servers and then merging the global model at the FIS. This causes a significant timing overhead and delays in data transmission. The CL strategy exclusively focuses on model training on a single-edge server. The presence of sequential data processing and limited computational resources results in significant bottlenecks, leading to a large increase in the training time, particularly when dealing with large-scale data. Therefore, our results suggest that, for the ISIC-2019 dataset, the mean latency 4.43 min, with a standard deviation of 2.09 min and a

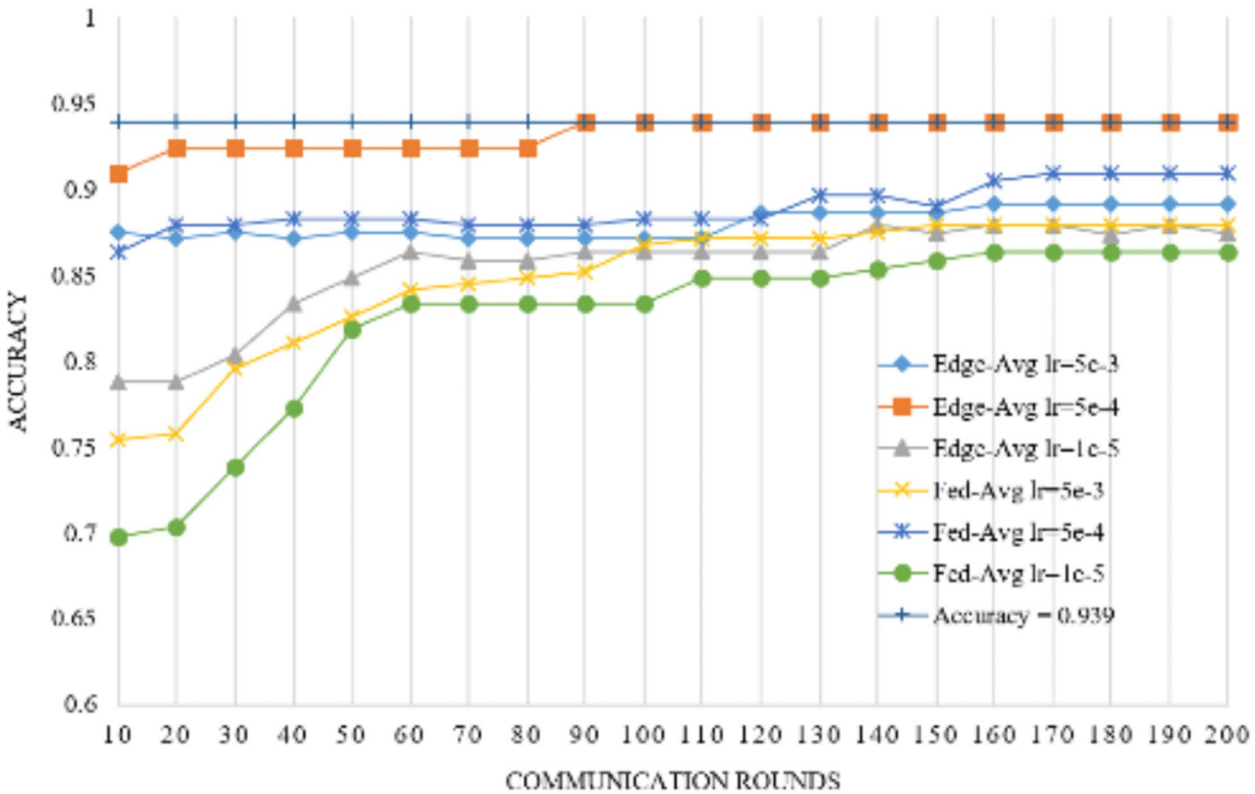


Fig. 5 The impact of communication rounds on accuracy

coefficient of variation of 47.17%. The Edge-Avg method demonstrated a 24.33% improvement over the Fed-Avg method and a 47.66% improvement over the CL method. In HAM10000 dataset, the mean latency was 13.25 min, with a standard deviation of 6.62 min and a coefficient of variation of 49.96%, where the Edge-Avg method achieved a 20.58% performance enhancement compared to the Fed-Avg method and a 47.69% enhancement over the CL method. For ISIC-2020, the mean latency was 42.84 min, with a standard deviation of 19.25 min and a coefficient of variation of 44.93%. The Edge-Avg method exhibited a 29.16% improvement over the Fed-Avg method and a 48.42% improvement over the CL method. This result establishes that our approach is a superior method for reducing the training delays in distributed learning.

Performance evaluation of Fed-Avg and Edge-Avg approaches

We evaluated the performance of Fed-Avg and Edge-Avg approaches, where accuracy is the secondary focus. The Edge-Avg approach improves Fed-Avg by addressing challenges from local data variations, such as differing image shapes, sizes, and techniques. These issues often bias results toward larger, higher-quality datasets, reducing accuracy on smaller or unique datasets and limiting adaptability. By using an adaptive strategy that emphasizes critical updates, Edge-Avg effectively handles local data differences. To evaluate this proposed method, we tested with different learning rates and examined the results of the communication rounds. In the case of Fed-Avg, the aggregation process occurs within the FIS, whereas for Edge-Avg, aggregation occurs on the edge server. As shown in Fig. 5, the accuracy analysis of the Fed-Avg under different learning rates revealed that the highest accuracy of 90.92% was achieved when utilizing a learning rate of 5e-4. Similarly, the Edge-Avg outperformed the Fed-Avg, attaining a significantly higher accuracy of 93.94% at the same learning rate. However, it is crucial to acknowledge that our trials revealed a decline in accuracy for both the Fed-Avg and Edge-Avg when the learning rate was beyond the ideal threshold. The accuracy dropped to 87.87% and 89.13% for the Fed-Avg and Edge-Avg, respectively, indicating the presence of overfitting and training process instability as contributing factors. Furthermore, we analyzed the loss

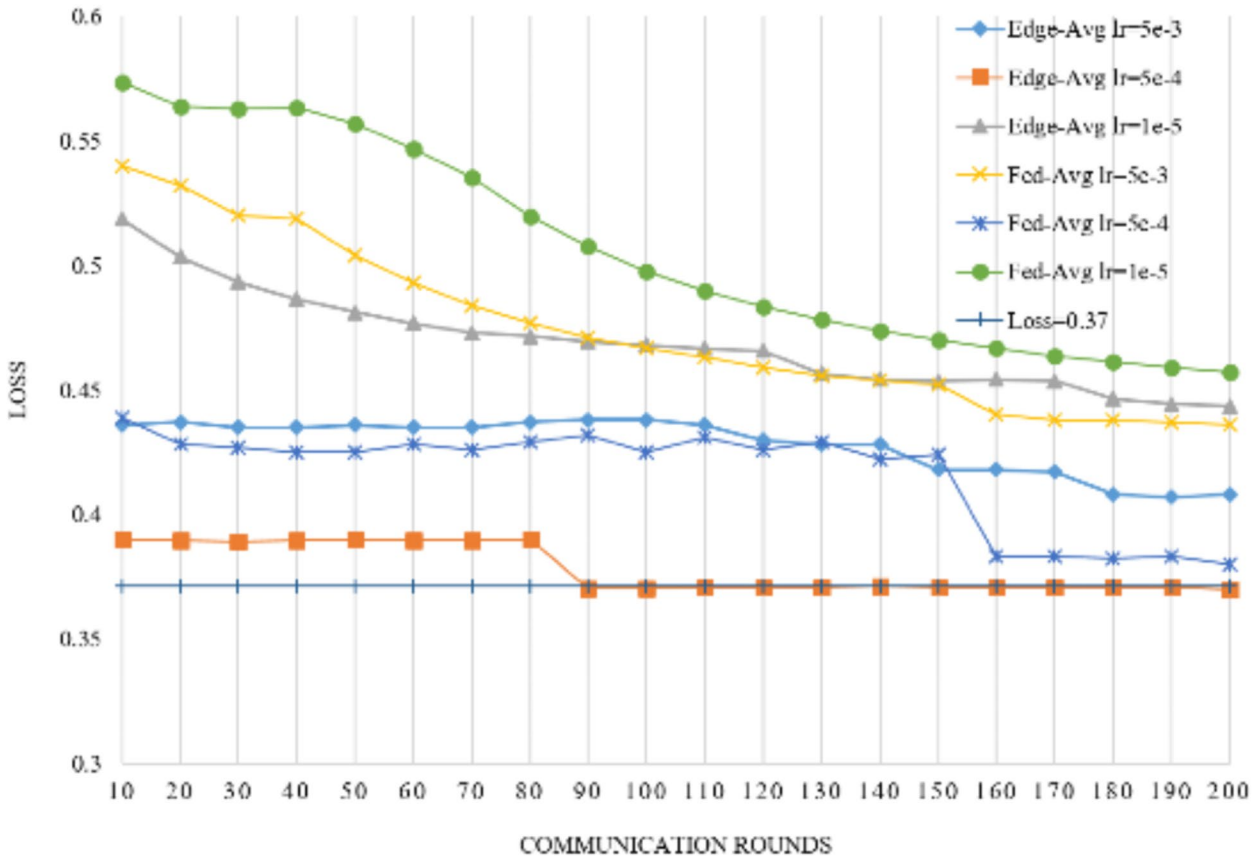


Fig. 6 The impact of communication rounds on loss

Table 2 Performance of Fed-Avg vs Edge-Avg

Method	*lr*	*Accuracy*	*Loss*	*Precision*	*Recall*	*F1-score*
Fed-Avg	5e-3	87.87	0.436	0.878	0.878	0.878
Fed-Avg	5e-4	90.92	0.380	0.909	0.909	0.909
Fed-Avg	1e-5	86.36	0.457	0.863	0.863	0.863
Edge-Avg	5e-3	89.13	0.408	0.891	0.891	0.891
Edge-Avg	5e-4	93.94	0.370	0.939	0.939	0.939
Edge-Avg	1e-5	87.39	0.443	0.873	0.873	0.873

Table 3 The proposed method is compared with cutting-edge models

Reference	Accuracy	Precision	Recall	F1-score
(2019) [35]	83.9%	0.709	0.56	-
(2020) [18]	86%	0.8	0.89	0.84
(2022) [20]	90.9%	-	0.886	0.896
Edge-Avg	93.94%	0.939	0.939	0.939

of Fed-Avg and Edge-Avg with different learning rates, as shown in Fig. 6. This study showed that a learning rate of 5e-4 achieved a minimum loss of 0.38. Similarly, Edge-Avg outperformed Fed-Avg in terms of loss, attaining a significantly lower value of 0.37 using the same learning rate. These results provide insights into the comparative performance of Fed-Avg and Edge-Avg, highlighting the potential benefits of the Edge-Avg method in terms of accuracy and loss reduction. Table 2 presents the detailed results.

Our edge-based federated learning approach has the potential to address the challenges faced by federated learning systems in edge-computing environments. Owing to limited resources and constrained communication bandwidth in such environments, leveraging the computational power of edge devices and performing learning tasks at the edge can reduce the communication overhead and improve system performance. The assessment of our method was conducted utilizing advanced deep learning metrics such as accuracy, loss, precision, recall, and F1-score, to deliver a comprehensive performance analysis highlighting the system's strengths and weaknesses. Our results on current data were compatible with the previous some research, for this ours approach increased accuracy and performance of edge-based distributed federated learning systems. As shown in Table 3. At the different domain such as healthcare and industrial automation, this edge computing approach can significantly impact real-time decision making.

Conclusion

This research examined edge intelligence training for developing a skin cancer detection system in healthcare facilities, with a particular focus on resolving challenges related to data privacy, delays in AI model training, and limitations in training data. Our thorough investigation unequivocally demonstrates that the edge intelligence training method substantially diminishes training delays compared to other well-known techniques, providing a more streamlined process for system development. In addition, our innovative Edge-Avg approach surpassed the Fed-Avg method in achieving better accuracy, highlighting its superior ability to handle complex data scenarios. The results strongly demonstrate that edge intelligence training is effective and efficient in creating resilient skin cancer detection systems in healthcare centers. In the future, our aim is to explore methodologies for enhancing the interpretability of federated learning models in medical image classification, including the utilization of explainable AI techniques within multi-access edge environments. Our approach offloads nonsensitive computations to central servers while processing sensitive data locally on edge servers. In addition, we design a reinforcement learning algorithms for dynamic resource allocation, workload optimization, and data heterogeneity.

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N/A.

Authors' contributions

Nithin Melala Eshwarappa contributed conceptual discussions and formulation, developing algorithms, coding the model, conducting simulation studies, obtaining results. Hojjat Baghban contributed in conceptual insight in terms of system model and algorithm conceptual discussion and formulation. Ching-Hsien Hsu contributed to conceptual discussions, actively participated in the literature review, and analyzed the results. Ren-Hung Hwang oversaw the findings, reviewed the content, and provided valuable feedback. Po-Yen Hsu and Mu-Yen Chen reviewed the work and offered insightful comments. All authors collectively discussed the results, finalized the manuscript, and approved the final version.

Data availability

In this research paper, we used the ISIC dataset [32], which is free dataset and public to use. The link address of the dataset is as follow: [32] Skin Cancer: Malignant vs. Benign — kaggle.com. https://www.kaggle.com/datasets/fanconic/skin-cancer-malignant-vs-benign . [Accessed 04-May-2024].

Declarations

Ethics approval and consent to participate
Not applicable.

Competing interests
The authors declare no competing interests.

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